A FOUR-QUBIT CLASSIFICATION EXPERIMENT

Quantum Wine Fingerprints

Can quantum similarities help me recognize wine cultivars?

178

wine samples

3

cultivar groups

4

logical qubits

The key question: does adding entanglement help?

METHOD

Give every model the same information

133 training wines / 45 held-out test wines / 3-fold training validation

13 measurements

Chemical composition
from the UCI Wine dataset

4 shared features

PCA compresses the data
for every model

3-group prediction

Classical models and
quantum-kernel SVMs

Quantum kernel = a similarity score between two encoded wines.

Choose settings on training data; use the test set only for evaluation.

Best test score: without entanglement

Correct predictions on the same 45 held-out wines



One fixed split. All quantum scores above were simulated on a classical computer.

45/45 on this test does not guarantee perfect predictions on new wines.

Classical methods are strong and quick

Measured time for hyperparameter search plus final training

0.15 s

Classical RBF SVM

2.61 s

Quantum: no entanglement

3.52 s

Quantum: entangled

Search grids differ, so these times include different numbers of candidates.
The kernel without entanglement is efficiently computable classically.

No quantum speedup or quantum advantage was demonstrated.

Real hardware changed two predictions

Pilot settings came from full training validation; this is a separate evaluation.

9/9

Local quantum + classical controls

7/9

IBM Fez hardware

186 circuits × 1,024 measurements each / 52 charged QPU seconds
Similarity error: 0.128 on hardware vs 0.059 with synthetic noise.
One job, nine test wines; no general hardware accuracy claim.

The simple noise simulation underestimated the measured hardware error.

A useful comparison, with clear limits

What this experiment supports — and what to investigate next

01 No entanglement benefit in this split

The model without entanglement had the best held-out score.

02 Classical controls remain essential

They were accurate and much faster to tune in this benchmark.

03 Hardware needs its own validation

Next: independent test splits and a budgeted error-reduction study.

Code, charts and raw evidence: github.com/MiroslawKmiec/Qff2026_heckathon

Data: UCI Wine · Aeberhard & Forina (1992) · DOI 10.24432/C5PC7J · CC BY 4.0